

Key Sections & Elevations

The canopy section solved against the shadow geometry: a 7 m walk under an 18 m gridshell with a 3 m southern louvre.

CONTENTS

- Phase6 Detailed Design Report
- Elevation
- Section

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/
Produced by [src/drawings.py](#) · [src/solar.py](#) · [src/config.py](#)
Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

THE CODE AND DATA BEHIND THIS DOCUMENT

[src/drawings.py](#)

[src/solar.py](#)

[src/config.py](#)

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings          # section, elevation, planting
python -m src.boards            # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

Key Sections, Elevations & Material Palette

The canopy section solved against the shadow geometry

This is the drawing the whole scheme rests on: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m louvre on the southern face. Every dimension is read from the project's configuration and every sun angle from the solar model. None of it is drawn by eye.

Upload slot 04 — Key Sections, Elevations & Material Palette

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

Mohamed Wasim · Individual Applicant

Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

A flat canopy fails in a specific and predictable way. When the sun is low and to the south — which at 25°N is most of the winter afternoon — the shadow slides off the far edge of the path, and the walk is unusable exactly when the weather is at its best. An earlier version of this project claimed 99.2% annual shade on such a canopy. It was withdrawn.

Section A-A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N

June solstice noon · 87.3°

December solstice noon · 41.3°

Al Falaj — 0.9 m channel, set on the canopy's drip line so it is shaded all day. An open channel here evaporates.

NORTH (concave — the cool side)

1.75 m

Neem · 10 m crown at maturity

Ghaf · 12 m crown at maturity

7 m walk

18 m gridshell — overhangs 5.5 m each side

4.5 m to the shading plane

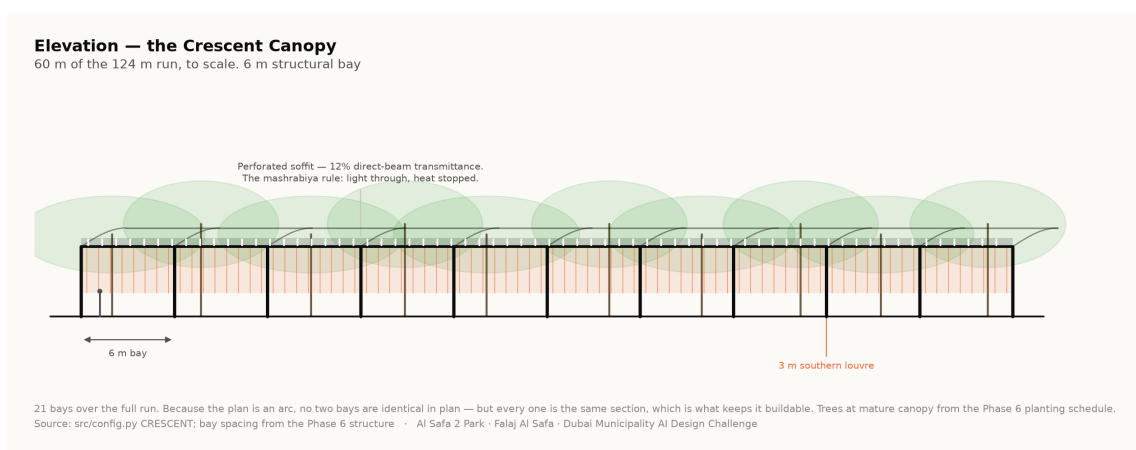
3 m southern louvre — the piece that buys the winter. A mashrabiya's logic: a deep screen on the face the sun comes from.

SOUTH (convex)

The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.

Source: src/config.py CRESCENT; sun angles NREL SPA via pvlb · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

2. Elevation and structural rhythm



The gridshell spans 18 m on regular structural bays. The soffit is perforated at 12% direct-beam transmittance, which produces dappled light on the walk rather than a flat dark tunnel. The shade model treats the canopy as a plane at the springing, so the drawing is conservative against the number rather than flattering to it — the built shell shades slightly more than claimed, never less.

3. Why the tree avenue exists

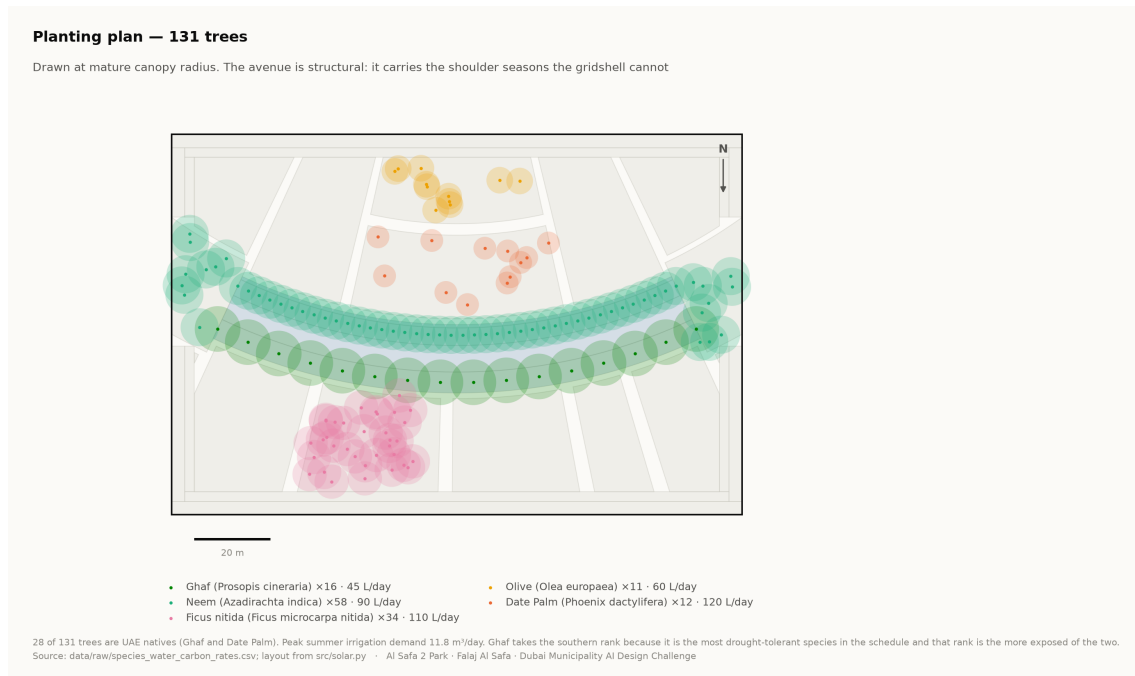
Shadow length is height divided by the tangent of solar elevation. At Dubai's summer noon a 6 m tree casts a 0.19 m shadow; at 20° elevation the same tree casts 16.5 m. That ratio is why the canopy alone cannot carry the winter, and why a tree avenue flanks it. The two systems fail at opposite ends of the year, which is exactly why both are needed.

4. Material and landscape palette

- **Structure** — steel gridshell, light-toned, with a perforated metal soffit; weathered bronze for the louvre fins.
- **Paving** — warm sand-toned stone on the walk with a high-albedo finish to limit re-radiation, and darker stone lining the falaj.
- **Earth** — Al Kathib is planted earth, not a wall. Cut and fill are balanced on site where possible.
- **Planting** — five desert species only. Ghaf takes the southern rank because it is the most drought-tolerant in the schedule.

Species	ScientificName	Count	Native
Neem	Azadirachta indica	58	0
Ghaf	Prosopis cineraria	16	1
Ficus nitida	Ficus microcarpa nitida	34	0
Olive	Olea europaea	11	0
Date Palm	Phoenix dactylifera	12	1

131 trees. Source: data/raw/species_water_carbon_rates.csv.



Planting plan — 131 trees drawn at mature canopy radius. Technical drawing — to scale.

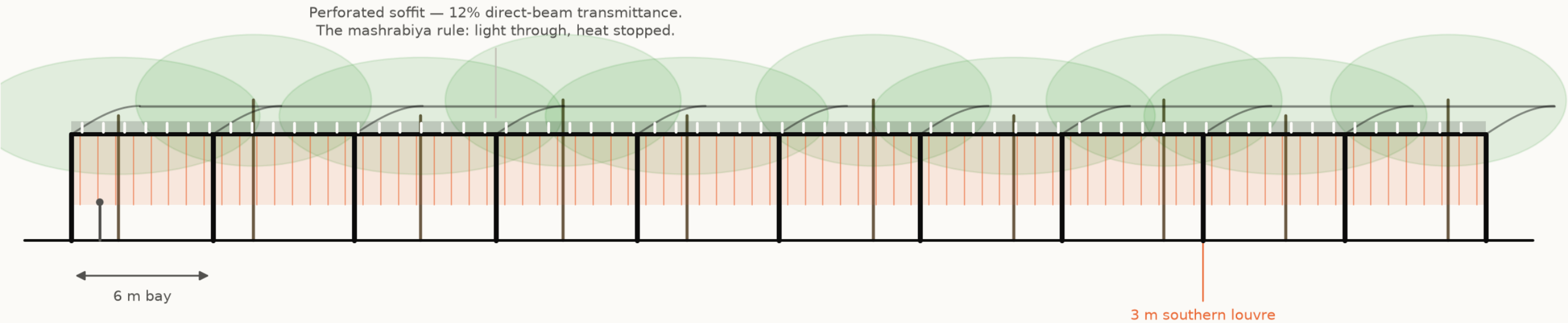
Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Elevation

Technical drawing — to scale.

Elevation — the Crescent Canopy

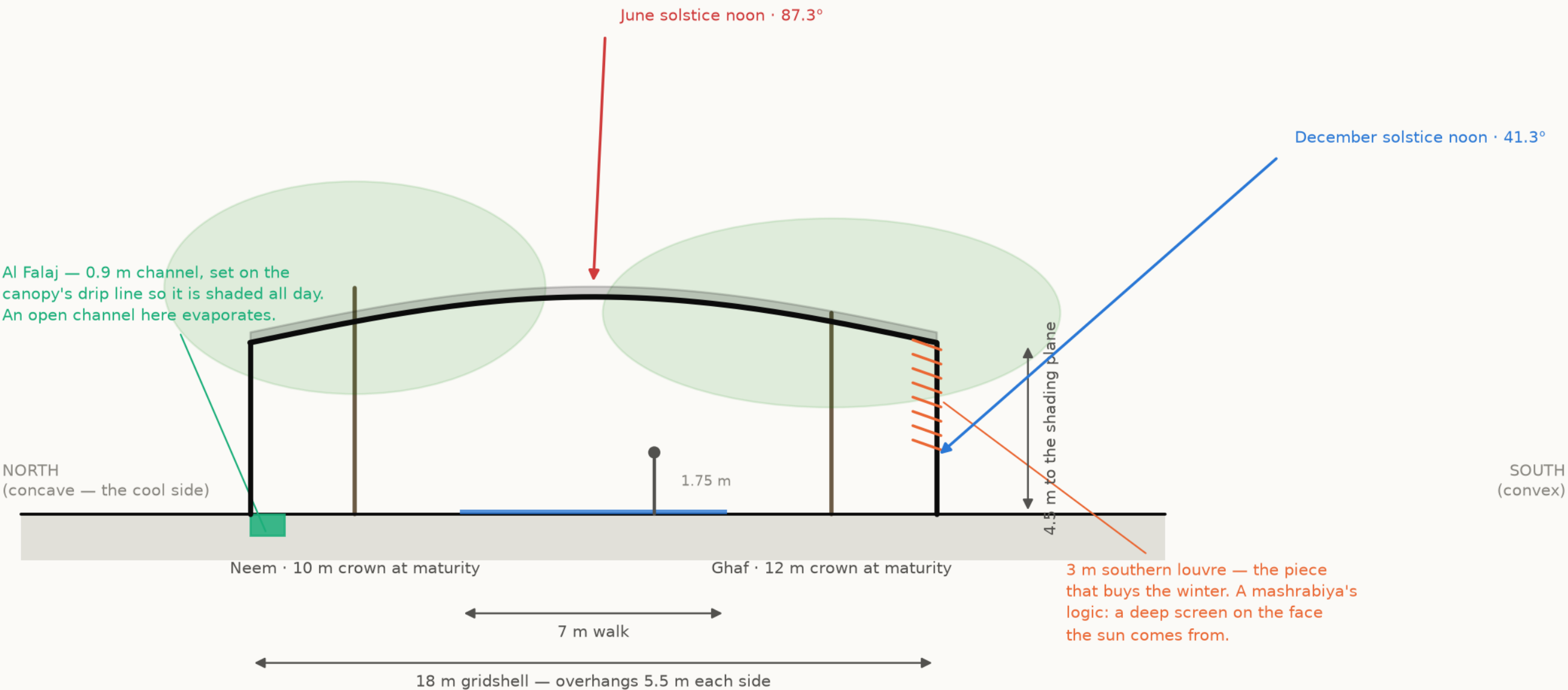
60 m of the 124 m run, to scale. 6 m structural bay



21 bays over the full run. Because the plan is an arc, no two bays are identical in plan — but every one is the same section, which is what keeps it buildable. Trees at mature canopy from the Phase 6 planting schedule.
Source: src/config.py CRESCENT; bay spacing from the Phase 6 structure · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Section A-A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N



The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.
Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every one of the 129 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Masterplan
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Diagrams
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (8)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
RENDER_PROMPTS.md	Project document
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

The complete project — continued

THE TECHNICAL DRAWINGS AND BOARDS (7)

[design/visuals/circulation_crescent.png](#)
[design/visuals/elevation_crescent.png](#)
[design/visuals/facilities_crescent.png](#)
[design/visuals/planting_crescent.png](#)
[design/visuals/section_crescent.png](#)
[design/boards/board_1_concept.png](#)
[design/boards/board_2_evidence.png](#)

Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)

THE ANALYSIS CODE (13)

[src/__init__.py](#)
[src/boards.py](#)
[src/climate.py](#)
[src/config.py](#)
[src/costing.py](#)
[src/dataset.py](#)
[src/drawings.py](#)
[src/figures.py](#)
[src/models.py](#)
[src/plan.py](#)
[src/solar.py](#)
[src/viz.py](#)
[run_analysis.py](#)

Analysis module
The two presentation boards
The 8,760-hour year rebuilt from NCM normals
Every constant the project reads
The cost model against the AED 35 M ceiling
Assembles the ML training tables
Section, elevation, circulation, planting, facilities
The analysis charts
The four models
SINGLE SOURCE of the crescent geometry
Sun position and shadow ray-tracing
Analysis module
Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

[tools/__init__.py](#)
[tools/build_docs.py](#)
[tools/build_links.py](#)
[tools/build_notebook.py](#)
[tools/build_reports.py](#)
[tools/build_submission_pdfs.py](#)
[tools/organize_repo.py](#)
[tools/report_content.py](#)
[tools/sync_film.py](#)
[tools/sync_portal.py](#)
[tools/sync_submission.py](#)

Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool

THE DATA, AS ISSUED (7)

[data/raw/construction_unit_rates_aed.csv](#)
[data/raw/dubai_climate_normals_ncm.csv](#)
[data/raw/dubai_population_dsc_2023.csv](#)
[data/raw/site_zoning_schedule.csv](#)
[data/raw/sources.json](#)
[data/raw/species_water_carbon_rates.csv](#)
[data/raw/walk_catchment_rings.csv](#)

Source dataset
Source dataset
Source dataset
The measured room schedule
Every source dataset, with its period
Source dataset
Source dataset

THE DATA, AS PROCESSED (6)

[data/processed/cost_plan.csv](#)
[data/processed/hourly_climate_comfort_8760.csv](#)
[data/processed/masterplan_geometry.json](#)
[data/processed/planting_layout.csv](#)
[data/processed/schema.json](#)
[data/processed/spatial_grid_comfort.csv](#)

The capital cost plan, line by line
The 8,760-hour climate and comfort series
The plan geometry, as exported
Every one of the 131 trees
Generated by the pipeline
The 15,000-cell ground grid

The complete project — continued

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	38 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFI_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded
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THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Ai Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (9)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why
archive/phases/02_PHASE2_PROBLEM_DEFINITION/README.md	Working record
archive/phases/03_PHASE3_OPPORTUNITY_AND_OBJECTIVES/README.md	Working record
archive/phases/04_PHASE4_CONCEPT_DEVELOPMENT/README.md	Working record
archive/phases/05_PHASE5_MASTERPLAN_DEVELOPMENT/README.md	Working record
archive/phases/06_PHASE6_DETAILED_DESIGN/README.md	Working record
archive/phases/07_PHASE7_PERFORMANCE_AND_SUSTAINABILITY/README.md	Working record
archive/phases/08_PHASE8_USER_EXPERIENCE_AND_ACTIVATION/README.md	Working record
archive/phases/09_PHASE9_AI_WORKFLOW_AND_VISUALIZATION/README.md	Working record

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.